

BRYAN WIESCHENBERG

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EDUCATION

The College of New Jersey (TCNJ), Ewing, NJ

August 2022 - May 2026

- **Major, BS:** Computer Science | **GPA:** 3.43/4.0 | **Honors:** Dean's List (multiple semesters)
- **Relevant Courses:** Software Engineering, Databases, Data Science, Computer Networking, Operating Systems

EXPERIENCE

Machine Learning Researcher, The College of New Jersey (TCNJ)

May 2025 - July 2025

- Engineered a Python data pipeline to augment real-world captures into a **40,000+** image training set, optimizing a proprietary AI vision model to improve accuracy by **50%**
- Architected a computer vision pipeline for spatial data extraction, achieving **30+** FPS by utilizing performance optimization strategies such as I/O multithreading, grayscaleing, and downsampling
- Integrated the pipeline with physical hardware to coordinate multiple robots, reducing manual intervention by **35%**

Operations Manager, The College of New Jersey (TCNJ)

August 2025 - Present

- Built a configurable Python-based scheduling engine to scrape availability data and apply weighted heuristics to generate optimized staff schedules with **100%** shift coverage, cutting scheduling time by **70%**
- Led a team of **15** staff in key inventory tracking, establishing strict data-handling protocols to support **3,000+** campus residents and reducing key distribution errors by **40%**
- Authored training guides and handoff documentation, reducing onboarding time by **30%** to increase new-hire velocity

PROJECTS

Research Paper Graph Visualizer ([Visit](#)) ([GitHub](#))

Feb 2026 - March 2026

- Visualized **200,000+** academic papers as an interactive, physics-simulated citation network using Next.js, TypeScript, Neo4j, and MongoDB, enabling researchers to explore cross-disciplinary connections
- Designed an optimized data pipeline achieving **8x** throughput gains, sustaining **200+** requests/sec under concurrent load and sub-**50ms** average latency via indexing, caching, and optimized queries
- Built an AI-powered graph analyst with Google Gemini, enabling natural-language queries to identify research clusters, knowledge gaps, influential papers, and cross-field connections across loaded datasets

Full-Stack Productivity Platform ([Visit](#)) ([GitHub](#))

Aug 2025 - January 2026

- Built and deployed a split-screen productivity platform on AWS EC2 with Next.js, TypeScript, and PostgreSQL, combining task management, calendars, and smart deadline alerts into a unified workspace
- Reduced average initial load latency by **65%** and cut API response times to sub-**30ms** under **50,000**-entity load through composite indexing, query refactoring, and prefetch pipelines with chunked pagination
- Architected a defense-in-depth auth system across **10+** security measures, including Redis rate limiting, reCAPTCHA, and OAuth 2.0 sign-in, hardening against SQL injection, XSS, brute-force, and credential stuffing attacks

End-to-End Encrypted LAN Chat App ([GitHub](#))

Jul 2025 - August 2025

- Designed and built a terminal-based LAN chat platform in Rust with client-side end-to-end encryption and persistent storage, sustaining **1,000+** concurrent connections at **~3ms** average round-trip time
- Engineered a modular role-based access control command dispatcher supporting **50+** commands, enabling room management, moderation, rate limiting, and session management directly from the chat interface
- Built a state-driven terminal UI engine supporting context-aware autocomplete, split panes, and dynamic redraws

SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, Rust, HTML, CSS
- **Technologies:** React, Next.js, FastAPI, Node.js, Express.js, PostgreSQL, MongoDB, Docker, AWS
- **Tools:** Cursor, Claude Code, Google Antigravity, GitHub Copilot, Git, Postman, cURL
- **Concepts:** REST APIs, system design, authentication, data modeling, threat modeling, CI/CD, secure software practices